

SUPPLEMENT TO NEURAL BELLMAN OPERATORS

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1. RELATION TO THE MAIN TEXT

This supplement expands the models and algorithms of the revised main manuscript. It does not introduce an alternative total loss or a different equilibrium definition. Policy evaluation freezes all actor outputs. Policy improvement freezes the critic's value and derivative channels. For games, only the acting player's objective is differentiated. For temporal selves, both exponential components evaluate the same future policy. The earlier main and supplementary sources are preserved in the revision archive for comparison; they are not alternative implementations of the revised method.

The inherited portfolio and quadratic laboratories retain their declared exact-structure classes. R6 additionally trains generic multilayer actors and critics in a stopped consumption economy, the NDU finite approximation, and dense nonlinear capital economies. The dynamic-game computation is a finite-grid best-response calculation. These approximation classes remain distinct throughout the run-level accounts below.

2. POLICY EVALUATION AND THE RECURSIVE POLICY GRADIENT

Let a_ϕ be differentiable in a scalar parameter direction, and suppose V^{a_ϕ} is sufficiently differentiable for the following calculation. The terminal payoff is independent of the policy parameter. Assume differentiation under the expectation is justified, that the differentiated state and utility equations are integrable, and that the boundary conditions of the differentiated problem are zero. Denote the directional derivative of the policy value by w .

Differentiating the fixed-policy equation gives

$$0 = w_t + \mathcal{L}^{a_\phi} w + f_y(t, x, a_\phi, V^{a_\phi}) w + D_a \mathcal{H}^{a_\phi}[V^{a_\phi}] D_\phi a_\phi.$$

The operator derivative includes the dependence of both drift and covariance on the action. The derivative of the generator acting on the changing value is contained in $\mathcal{L}^{a_\phi} w$, while the action derivative of the generator acting on the fixed value is contained in $D_a \mathcal{H}$. No Hessian-action channel is discarded.

Applying the linear Feynman–Kac formula to the differentiated equation gives

$$w(t, x) = \mathbb{E}_{t,x}^{a_\phi} \int_t^\tau \exp \left(\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv \right) D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds.$$

This formula concerns the performance derivative from the stated initial state. A domain-uniform Hamiltonian objective uses a different weighting. Such an objective is useful for approximate pointwise improvement, but it becomes a lifetime policy gradient only with the appropriate occupation and adjoint weights. Trajectories sampled by the current actor are detached in the implemented domain objective. Differentiating through their sampling law would define a different gradient estimator and must be analyzed separately.

The critic’s residual derivative is the derivative of the entire fixed-policy equation. In particular, its level is determined by the recursive evaluation equation together with boundary data. A separate negative-Hamiltonian gradient in the critic is not an innocuous level anchor: in the scalar stationary example it moves the minimizer away from the Bellman root.

3. REPRESENTATION, BOUNDARY CONDITIONS, AND ERROR MEASUREMENT

3.1. Smooth approximation and what it establishes

A derivative-approximation result concerns a smooth target on a compact domain and a family of networks whose capacity can increase. It does not imply that a fixed network attains zero residual, that optimization finds its best approximation, or that a nonsmooth target has classical derivatives everywhere. In a smooth problem, increasing the representation class can reduce approximation error in the value and derivatives. In a nonsmooth problem, NBO instead evaluates a consistent positive-weight Bellman equation, or solves an explicitly regularized equation before taking a justified limit.

The hard-terminal architecture is $v(t, x) = G(x) + (T - t)N(t, x)$. It exactly enforces $v(T, x) = G(x)$ independently of training. A lateral Dirichlet face needs an additional compatible lifting and vanishing factor. For a reflected state, the Neumann condition is a different constraint. A loss on only interior points leaves all of these boundary questions unanswered. The regression test evaluates the terminal architecture at randomly chosen states and checks exact equality, not merely a small mean terminal error.

The recursive Epstein–Zin domain is $bV > 0$. An exponential output ensures positivity, while an interval transformation can keep bV between declared positive barriers. If $bG > 0$, one terminal-compatible positive representation is

$$bv(t, x) = \exp [\log(bG(x)) + (T - t)N(t, x)].$$

Its terminal value is exact. It does not by itself impose a lateral exit payoff or a two-sided barrier. These additional conditions must be built into the architecture or measured explicitly. Clipping an invalid utility argument inside a fractional power changes the equation and is not an acceptable hidden numerical repair.

3.2. From residual samples to a uniform bound

Let r be K -Lipschitz on a domain in $\mathbb{R}^d$. If every point is within distance h_V of a validation point, the triangle inequality gives $\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kh_V$. If K is estimated from samples rather than bounded, the resulting expression is itself a diagnostic, not a certified uniform bound.

An integral bound needs more assumptions. Suppose the sampling density is at least $p_0 > 0$, and each ball of radius r_0 or less centered in the domain intersects the domain in volume at least $c_D r^d$. Let $M = \|r\|_\infty$ and assume $M/(2K) \leq r_0$. Near a point approaching the maximum, Lipschitz continuity gives $|r| \geq M/2$ on a ball of radius $M/(2K)$. Hence

$$\mathbb{E}r^2 \geq \frac{p_0 c_D}{2^{d+2} K^d} M^{d+2}, \quad M \leq \left(\frac{2^{d+2} K^d \mathbb{E}r^2}{p_0 c_D} \right)^{1/(d+2)}.$$

For larger M , use the capped radius r_0 and the corresponding two-case inequality. A concentration argument is additionally needed to replace the population mean by

an empirical one. This calculation explains why a dimension-free uniform certificate cannot be read directly from an average training loss.

The action gap is treated similarly. If an action net has covering radius h_A and the Hamiltonian is K_A -Lipschitz in the action, the unobserved supremum exceeds the net maximum by at most $K_A h_A$. Thus the net gap plus this term is an upper bound. A collection of successful deviations without a covering or duality argument is only a lower bound on exploitability.

4. BENCHMARK DERIVATIONS

4.1. Merton consumption and transversality

Write $b = 1 - \gamma$ and $V = KX^b/b$. Then $V_X = KX^{-\gamma}$ and $V_{XX} = -\gamma KX^{-\gamma-1}$. Substituting $c = mX$ into the stationary time-additive equation and dividing by the positive number KX^b gives

$$0 = \frac{m^b}{bK} - \frac{\rho}{b} + r + p(\mu - r) - m - \frac{1}{2}\gamma p^2 \sigma^2.$$

Consumption maximization gives $m^{-\gamma} = K$; portfolio maximization gives $p = (\mu - r)/(\gamma\sigma^2)$ when interior. With a no-short-sale constraint and negative excess return, the maximizing share is zero. Let $R_* = r + p(\mu - r) - \gamma p^2 \sigma^2 / 2$. After substitution, $\gamma m = \rho + (\gamma - 1)R_*$.

For the optimal constant shares, $e^{-\rho t} \mathbb{E}[X_t^{1-\gamma}]$ decays as e^{-mt} . Thus $m > 0$ gives the candidate's additive transversality condition. Verification against a larger policy class also requires the relevant localization and integrability conditions for the competitors, as specified in the main text. Policies with minus-infinite utility are not preferred to the finite candidate. The numerical laboratory checks the candidate formulas and does not silently apply a stationary constant to an unspecified finite-horizon terminal problem.

4.2. Recursive coefficient and the utility interval

For $f(c, V) = \rho[c^a(bV)^{1-a/b} - bV]/a$, the same homothetic substitution gives

$$0 = \frac{\rho}{a} [m^a K^{-a/b} - 1] + R_* - m.$$

Since $f_c = V_X$ implies $\rho m^{a-1} K^{-a/b} = 1$, the first term becomes $(m - \rho)/a$. Consequently, $m = \psi\rho + (1 - \psi)R_*$ and $K = [\rho m^{a-1}]^{b/a}$. The formula assumes the stationary candidate has positive consumption. It is not an infinite-horizon utility-selection theorem.

For the finite-horizon matching-bequest test at $a > 0 > b$, define $Y = bV$. The consumption domain is compact and bounded away from zero. At sufficiently small positive Y , $c^a Y^{-a/b} < 1$ uniformly in consumption, so $f < 0$. At sufficiently large Y , the reverse inequality holds and $f > 0$. Because dividing by negative b reverses the order of the utility endpoints, the large- Y constant is a lower utility barrier and the small- Y constant is an upper utility barrier. The matching bequest lies between them after enlarging the interval. Lipschitz scalar comparison then keeps each policy utility inside the interval, on which the aggregator and its utility derivative are bounded. This verifies the finite-horizon interpretation without claiming an unrestricted infinite-horizon result at those parameters.

5. ENDOGENOUS-PREFERENCE DETAILS

5.1. State restrictions and the reference experiment

In the baseline application, the risk-aversion interval lies strictly above one and the process is reflected. The reference wealth problem is stopped at positive finite bounds. Consumption shares are bounded away from zero, adjustment efforts are bounded, and the horizon is finite. These choices bound the flow utility, its relevant derivatives away from the terminal-boundary compatibility issue, and the bequest. They provide a well-defined finite reference calculation. The numerical reflection is a folded reflection of Brownian cubature endpoints; the wealth boundary is absorbing with the prescribed payoff. The same terminal and boundary values are imposed for both adjustment-cost parameters.

The grid reference operates in $y = \log X$. Its drift is

$$b_y = r + p(\mu - r) - m - \frac{1}{2}p^2\sigma_X^2,$$

and its covariance with u is $\rho\sigma_u p\sigma_X$. The two Brownian-factor columns can be written $(\sigma_u, \rho p\sigma_X)^\top$ and $(0, \sqrt{1 - \rho^2} p\sigma_X)^\top$. Four symmetric cubature directions re-

1 produce this covariance. The correction $-p^2\sigma_X^2/2$ in log wealth is essential; omitting 1
 2 it would change the economic diffusion. 2

3 For each action, the backward update is current utility times the time step plus 3
 4 the exponentially discounted positive-weight interpolation of next-period value. 4
 5 Feasible actions are enumerated on the declared 27-element grid. Because the 5
 6 terminal/exit payoff is fixed, raising k weakly lowers each action's payoff and, by 6
 7 backward induction, lowers value at every state and time. The recorded arrays 7
 8 permit a direct check of this order. The reported grid-refinement difference is not an 8
 9 estimate of unrestricted neural approximation error. 9

12 5.2. Policy conditions and comparative statics 12

14 The consumption marginal utility in reference units is $U_c = \bar{c}^{-1}(c/\bar{c})^{-u}$. Equating 14
 15 it to V_X yields $c = \bar{c}(\bar{c}V_X)^{-1/u}$. Adjustment maximizes $\theta V_u - k\theta^2/2$ and hence 15
 16 uses the projection of V_u/k under a box. Portfolio choice maximizes a quadratic 16
 17 with coefficient $X^2\sigma_X^2 V_{XX}/2$. The projected first-order solution is valid only when 17
 18 this coefficient is negative. These conditions clarify why numerical curvature and 18
 19 feasibility must be checked independently. 19

20 The cross derivative V_{uX} is an endogenous object. Its sign cannot be inferred from 20
 21 the label "risk aversion" or from a negative shock correlation. Nor can the sign of 21
 22 $\partial_k \theta^k$ be inferred by differentiating $1/k$ while keeping V_u^k fixed. In contrast, the value 22
 23 monotonicity in k holds policy by policy. At a differentiable optimum the envelope 23
 24 formula follows because k appears only in the quadratic running cost. Changing 24
 25 the normalization function $\kappa(u)$ can change both adjustment and portfolio policies 25
 26 even while leaving utility curvature in consumption unchanged. 26

27 An observation model is needed for structural estimation. With latent u , uncer- 27
 28 tainty in κ , and unknown adjustment or shock parameters, a sensitivity plot for 28
 29 one policy component does not prove identification. A practical estimation exercise 29
 30 would compute moment derivatives jointly, inspect their rank after normalization 30
 31 restrictions, and propagate numerical approximation error into the implied moment 31
 32 map. Those are distinct tasks from solving the controlled economy. 32

6. TEMPORAL-SELF DEVIATIONS

For the two-exponential kernel, the component value under constant consumption share m is $W_j(X) = m^b X^b / [b\lambda_j(m)]$, where $\lambda_j(m) = \rho_j - b(r - m)$. A deviation to share z for duration ϵ , followed by m , has component value at $X = 1$

$$J_j(z, \epsilon) = \frac{1}{b} \left[z^b \frac{1 - e^{-\lambda_j(z)\epsilon}}{\lambda_j(z)} + e^{-\lambda_j(z)\epsilon} \frac{m^b}{\lambda_j(m)} \right].$$

The continuous extension is used if $\lambda_j(z) = 0$. The deviation gain is $\sum_j w_j J_j(z, \epsilon) - \sum_j w_j W_j(1)$, with weights $(\beta, 1 - \beta)$. This is the quantity independently maximized in the experiment, rather than a restatement of the equilibrium first-order condition.

At an equilibrium share, the first-order gain is nonpositive for every feasible instantaneous share and equals zero at the equilibrium share. A finite-duration self can generally choose a slightly different share, so the maximized gain need not be exactly zero for positive duration. The required diagnostic is its gain divided by duration as duration tends to zero. For $\beta = 1$ the ordinary exponential solution is recovered.

As κ grows, the kernel becomes increasingly concentrated near the present before approaching its long-run component. Taking the instantaneous-deviation limit and taking $\kappa \rightarrow \infty$ are conceptually different operations. The finite- κ model in this paper explicitly specifies the order: the spike condition is derived for fixed κ , and any duration-limit comparative static is taken afterward. This avoids assigning an undefined continuous-time meaning to a discrete present-bias multiplier.

7. GAMES AND INDEPENDENT BEST RESPONSES

A fixed rival profile turns each player's problem into an ordinary controlled Markov problem, so the main verification results apply player by player. The derivative graph must implement this reduction. During player i 's actor step, every rival policy and every derivative of player i 's critic is detached. During player i 's critic step, all players' actions are detached. A sum of these reporting losses can be logged, but differentiation of a joint-profit objective is not equivalent to those block steps.

The two-firm static diagnostic has best response $q_i = (1 - q_j)/2$ when interior. At the joint-profit allocation $(1/4, 1/4)$, the deviating firm chooses $3/8$ and improves profit from $1/8$ to $9/64$, a gain of $1/64$. The own-action derivative is $1/4$ at the joint optimum. These numbers verify a graph-level distinction; they are not a computation of the dynamic capital game.

For the dynamic problem, a reported equilibrium should include an independent solution of each fixed-rival best-response problem, or an upper bound on the unilateral Hamiltonian gap over the entire verification domain. A finite list of unsuccessful deviations does not supply such an upper bound. Multiple initializations, asymmetric profiles, boundary actions, and the market-size normalization are part of the same experiment. The specification in the main text retains the dynamic Cournot application but does not invent a dynamic Nash table from the static arithmetic.

8. STOCHASTIC HESSIAN OBJECTIVES AND COMPLEXITY

Let $B = \Sigma^\top D^2 v \Sigma$, a symmetric matrix in Brownian-factor space. For Gaussian probes, Isserlis' identity gives $\mathbb{E}(z^\top B z) = \text{tr } B$ and $\text{Var}(z^\top B z) = 2 \text{tr}(B^2) = 2 \|B\|_F^2$. Averaging K independent probes divides the variance by K . Rademacher probes instead have variance $2 \sum_{i \neq j} B_{ij}^2 / K$. The variance depends on the chosen covariance factor and probe law, both of which must be specified.

Write $\hat{q}_j = q + \epsilon_j$, with conditionally mean-zero independent errors. Then

$$\mathbb{E}[(d - \hat{q}_1/2)(d - \hat{q}_2/2)] = (d - q/2)^2 + \frac{1}{4} \mathbb{E}[\epsilon_1 \epsilon_2] = (d - q/2)^2.$$

Using one bank twice would retain its variance term. In the scalar Gaussian example, a single-bank squared residual has expectation $1 - h + 3h^2/4$, minimized at $h = 2/3$, whereas the exact residual $(1 - h/2)^2$ is minimized at $h = 2$. The independent product restores the latter population objective. The saved Monte Carlo diagnostic uses 400,000 pairs and records both empirical estimates; it does not claim that the sampling error is identically zero.

The core implementation uses exact Hessians to make the covariance contraction independently testable. A Hessian-vector implementation can replace it without changing the conditional-independence requirement. The cost of a Hessian-vector

product depends on the network graph, not just the state dimension. Total cost also includes sampling, optimization, domain coverage, action maximization, and stopping at an agreed error level. A quadratic-neural representation with a Lyapunov solve has different approximation and optimization costs from a generic multi-layer network. The inherited quadratic laboratory is explicitly the former; the new dense nonlinear experiments actually train generic networks and Hessian-vector estimators.

9. REPRODUCTION AND FURTHER COMPARISONS

From the repository root, execute the experiment program and then the regression suite. The dependency versions and CPU setting are recorded with the results. The experiment output distinguishes acceptance of a calibration or identity test from execution of a reference computation. A separate reviewer-check output reproduces the original twelve diagnostics against the archived original manuscript, not against the revised article.

A broader method comparison should use one problem specification and evaluate value error, policy error, boundary error, and the independent action gap at matched accuracy. A direct neural-HJB comparator eliminates the actor with the same feasible maximization routine. A temporal-regression comparator uses the same recursive discretization and utility domain, without adding discount twice. A separability-aware classical comparator exploits all available economic structure. Network capacity, sampling, covariance treatment, random seeds, failures, wall time, and memory should all be reported. R6 executes the architecture-matched direct-HJB and finite-Bellman comparisons. The broader deep-BSDE and neural-game comparisons remain distinct unexecuted alternatives; they are not counted as measured advantages.

The inherited read-only workflow is preserved. The separately scoped R6 replication workflow commits materialized source and raw recomputation outputs only to its designated candidate branch, without editing the review or earlier revision branches. Hashes and exact source commits distinguish these artifacts. No workflow rewrites a failed numerical accuracy target as a pass.

10. IMPLEMENTED ERROR ACCOUNTS AND ARITHMETIC

The implemented map returns a candidate and a verification record, rather than assigning authority to an optimizer's exit code. In the differential consumption economy, the record identifies the binary64 weights, both stopping faces, all dyadic leaves, global consumption maximization, and the error bounds for the continuous diffusion. In NDU it instead identifies the finite state–action–time model, its positive transition weights, and time-indexed evaluation and improvement errors. The dynamic-game record fixes the entire rival policy before an independent backward best-response solve. An unchecked continuous-action or time-discretization term is represented as unavailable in the machine-readable output, not as zero.

For finite time levels, let $\delta_t = \|v_t - T_t^a v_{t+1}\|_\infty$ and $\epsilon_t = \|T_t v_{t+1} - T_t^a v_{t+1}\|_\infty$. If both maps are β_t -Lipschitz, the triangle inequality gives

$$\|v_t - V_t^a\| \leq \delta_t + \beta_t \|v_{t+1} - V_{t+1}^a\|, \quad \|v_t - V_t^*\| \leq \delta_t + \epsilon_t + \beta_t \|v_{t+1} - V_{t+1}^*\|.$$

Summing the two inequalities yields the regret account. The same recurrence with a player's unilateral gap bounds the gain from a dynamic deviation when that player's policy evaluation is exact. Accounts must be checked at each time level: a time-zero bound need not dominate a later-time bound when discounting and errors differ. The coarse-game regression exercises this distinction and verifies that a missing pure equilibrium is reported rather than silently treated as exact.

10.1. Full-domain interval implementation

An interval $[l, u]$ encloses an exact real quantity. Each elementary operation expands its rounded endpoints with the adjacent binary64 number. Affine layers split positive and negative weights and include a conservative dot-product forward-error allowance. The arithmetic contract requires binary64 round-to-nearest arithmetic, the stated dot-product error bound, and absence of exceptional overflow or flush-to-zero behavior. Stored parameters denote their exact binary64 real values; the verifier does not assert an unrecorded decimal model identity.

For an exponential argument x , range reduction gives $r = x/2^s$ with $|r| \leq 1/8$. A degree-fourteen Taylor polynomial has remainder bounded by

$$e^{|r|} |r|^{15}/15! < 2(1/8)^{15}/15! < 2^{-80}.$$

Interval Horner evaluation and repeated squaring enclose e^x . The identity $\tanh x = 1 - 2/(e^{2x} + 1)$ then supplies $\tanh$ values and derivative channels $1 - \tanh^2 x$ and $-2 \tanh x(1 - \tanh^2 x)$. Powers of two and the range-reduction domain are checked in the code.

For logarithms, exact binary decomposition gives $x = 2^k m$, with $1 \leq m < 2$. Writing $z = (m - 1)/(m + 1)$,

$$\log m = 2 \sum_{j=0}^{24} \frac{z^{2j+1}}{2j+1} + R, \quad |R| \leq \frac{2|z|^{51}}{51(1-z^2)} < 2^{-72}$$

for the outward-enclosed range $|z| < .334$. The same expansion encloses $\log 2$. Neither construction presumes correctly rounded library $\tanh$ or logarithm functions. The saved test suite checks transcendental endpoints against eighty-digit arithmetic, differentiates the trained network independently, and checks all certificate hashes and leaf adjacencies. These tests supplement, but do not replace, the arithmetic contract.

The scalar action objective $h(c) = \log c - pc$ has second derivative $-1/c^2$. If C encloses both the actor and its maximizing alternative, strong concavity on their joining segment gives

$$h(c^*) - h(c) \leq h'(c)(c^* - c) - (c^* - c)^2/(2C^2) \leq C^2 h'(c)^2/2.$$

The argument includes constrained endpoints. Taking the minimum with the natural interval action-gap enclosure markedly reduces verification cost, without using a sampled gradient as a bound. Residual targets are decided on every leaf; reaching a maximum subdivision depth does not change a failed target into a pass.

10.2. Application-specific hypotheses

The stopped consumption diffusion is uniformly nondegenerate on its compact domain; controls and coefficients are bounded and its endpoint data are continuous.

The smooth networks and strictly positive consumption map meet the regularity and log-domain requirements of the verification argument. Both stopping-face errors and the full action interval are checked. The upwind reference error is not a premise of its continuous certificate.

The NDU computation is a positive-weight, finite-action, finite-horizon economy with prescribed discrete stopping and folded preference transitions. Positivity, all terminal and wealth-face values, and every finite action are evaluated. The continuous reflected PDE's corner compatibility, continuous-action coverage error, and diffusion approximation error are not inferred from these tests. Expanding a feasible box changes the constrained economy; densifying a fixed box does not. The sensitivity record keeps these operations separate.

The dense capital computation uses bounded nonlinear drift in log coordinates, bounded positive consumption fractions, constant diffusion, and finite horizon. Its flow has at most linear log-state growth and its terminal payoff at most quadratic growth; these are integrable under the declared bounded-drift diffusion. The fitted box is a sampling domain, not a state constraint. Its sampled differential residuals do not check comparison barriers or uniform errors on $\mathbb{R}^d$. Euler payoff estimates carry both sampling error and unbounded-state/time-discretization issues not covered by the reported Monte Carlo standard error.

The dynamic capital game is a fully enumerated finite approximation with stopped joint boundaries. Stage equilibria and dynamic fixed-rival best responses are numerical statements about that model. The continuous-control best-response comparison assumptions in the main theorem remain logically distinct. The retained homothetic recursive-utility and temporal-self examples use their stated utility domains, bequests, and integrability conditions; R6 does not relabel them as unrestricted neural experiments.

11. DESIGN, BUDGETS, AND REPRODUCIBILITY

The stopped-consumption study uses float64 and the existing generic derivative core. Identical seed labels fix initialization and collocation construction across methods. L-BFGS iteration caps are matched, but line-search closure counts need not match. Those counts, fit times, and separate verification times are saved. The

1 grid reference uses exact maximization within each upwind drift region, not the 1
2 learned actor. Reference comparisons omit four endpoint-adjacent nodes only for 2
3 finite-difference derivative comparisons; all endpoints enter the boundary check 3
4 and the interval certificate. 4

5 The NDU actor predicts one of the finite feasible triples. Its normalized Bellman- 5
6 loss objective gives economically costly action mistakes more weight than ordinary 6
7 classification would. Absorbing boundary rows are excluded only from actor fitting, 7
8 because their actions have no effect; they remain in policy evaluation and verifi- 8
9 cation. The failed initial classification implementation is archived with its actual 9
10 outputs. Principal seed 11 misses the declared policy-regret target even after the 10
11 improvement; its result is not discarded. The direct finite-Bellman comparator uses 11
12 the same critic class and total inner-iteration cap. 12

13 The dense-capital primary study uses exact Hessians; the trace follow-up actu- 13
14 ally backpropagates through Hessian-vector products. Product and single-bank 14
15 implementations use four probes in total, but only the product splits them into 15
16 conditionally independent banks. The Monte Carlo regression separately checks 16
17 the value and gradient of the product objective with tolerances .012 and .025. An 17
18 unbiased objective can have higher variance, and no claim of superior finite-sample 18
19 accuracy follows merely from its expectation. The covariance in the retained experi- 19
20 ment is modest; the reported results reflect that scale. 20

21 The initial high-dimensional state box was too narrow to certify the realized 21
22 policy paths. The wider-box design is therefore identified as a follow-up, not a 22
23 preregistered experiment. Forty- and eighty-step policy simulations share the same 23
24 seed convention across methods at a given resolution. They are not a coupled 24
25 Brownian extrapolation across resolutions. Raw per-path payoffs allow paired 25
26 policy comparisons, but neither a Monte Carlo confidence interval nor a critic- 26
27 payoff agreement proves closeness to the optimal value. 27

28 Each executed program records all declared seeds, its budget, residual or devia- 28
29 tion arrays, and an explicit scope. Process peak resident memory is a high-water 29
30 mark of the process, potentially including earlier runs. It is not an isolated memory- 30
31 complexity measurement for an individual method. CPU timings are execution- 31
32 specific and do not reproduce the historical accelerator claims. The independent 32

raw-output workflow binds its source commit and file hashes; the source archive keeps the R2 manuscript and supplement byte-for-byte for comparison.

12. RUN-LEVEL NUMERICAL ACCOUNTS

12.1. Consumption fitting and verification

Method	Seed	Width/depth	Fit seconds	Verify seconds	Pass
nbo	11	16/2	2.1790	1.0822	True
nbo	29	16/2	2.0422	1.4045	True
nbo	47	16/2	2.0550	1.3100	True
direct	11	16/2	1.9767	0.4974	True
direct	29	16/2	0.8018	0.4875	True
direct	47	16/2	2.1912	0.5481	True
joint	11	16/2	0.2578	–	False
joint	29	16/2	2.5836	–	False
joint	47	16/2	2.5935	–	False
nbo	11	8/2	2.5767	–	True
nbo	11	32/2	1.6599	–	True
nbo	11	16/1	1.6198	–	True
nbo	11	16/3	2.3456	–	True
nbo	11	16/2	2.1977	–	True
nbo	11	16/2	1.8599	–	True

Pass in this table means the prespecified sampled diagnostic. Continuous certification is reported separately and only for the primary NBO and direct-HJB runs. Every closure count, pointwise derivative array, parameter, and iteration trace is in the corresponding raw result.

12.2. Action bounds and numerical domains

Actions	Grid/time	k	Center value	Upper m freq.	Upper p freq.
r2	17x25/20	1.0000	-11.9152	0.9994	0.7885
r2	17x25/20	5.0000	-11.9570	0.9992	0.7890

Actions	Grid/time	k	Center value	Upper m freq.	Upper p freq.
adjustment	17x25/20	1.0000	-10.6081	1.0000	0.8364
adjustment	17x25/20	5.0000	-10.9202	1.0000	0.8405
expanded	17x25/20	1.0000	-4.8410	0.9898	0.5440
expanded	17x25/20	5.0000	-4.9243	0.9855	0.5515
refined	17x25/20	1.0000	-4.8406	0.9737	0.5301
refined	17x25/20	5.0000	-4.9214	0.9688	0.5353
expanded_domain	17x25/20	1.0000	-2.7504	0.9698	0.5665
expanded_domain	17x25/20	5.0000	-2.8452	0.9499	0.6068
expanded	17x25/20*	1.0000	-5.0185	0.9997	0.3164
expanded	17x25/20*	5.0000	-5.1095	0.9971	0.3251
expanded	25x37/40	1.0000	-4.8395	0.9902	0.5738
expanded	25x37/40	5.0000	-4.9228	0.9853	0.5787
expanded	33x49/80	1.0000	-4.8302	0.9872	0.6050
expanded	33x49/80	5.0000	-4.9143	0.9823	0.6102

An asterisk marks wealth interval $[\cdot 25, 4]$, rather than $[\cdot 5, 2]$. Frequencies use all nonabsorbing state–time nodes. The raw record includes both bounds for each of the three controls. “Expanded has 175 actions; “refined has 1,053 actions in the same box; “expanded_domain changes the box itself.

12.3. Dense neural runs, trace estimators, and coverage

Method	d/seed	Steps	RMS	Gap	Seconds	Pass
exact	2/11	600	0.0125	1.44×10^{-4}	2.7329	True
direct	2/11	600	0.0123	0	3.4870	True
exact	2/29	600	0.0161	2.28×10^{-4}	2.6554	True
direct	2/29	600	0.0161	0	3.4759	True
exact	2/47	600	0.0132	2.42×10^{-4}	2.6361	True
direct	2/47	600	0.0136	0	3.5572	True
exact	5/11	600	0.0101	0.0012	4.1518	True
direct	5/11	600	0.0101	0	5.0391	True
exact	5/29	600	0.0131	8.38×10^{-4}	4.2150	True
direct	5/29	600	0.0129	0	4.9992	True

	Method	d/seed	Steps	RMS	Gap	Seconds	Pass	
1								1
2	exact	5/47	600	0.0122	0.0020	4.1258	True	2
3	direct	5/47	600	0.0118	0	5.0485	True	3
4	exact	10/11	600	0.0163	0.0052	6.8652	True	4
5	direct	10/11	600	0.0138	0	7.6139	True	5
6	exact	10/29	600	0.0188	0.0035	6.9393	True	6
7	direct	10/29	600	0.0168	0	7.7251	True	7
8	exact	10/47	600	0.0141	0.0050	6.8241	True	8
9	direct	10/47	600	0.0139	0	7.6988	True	9
10	exact	20/11	600	0.0148	0.0113	12.8180	False	10
11	direct	20/11	600	0.0135	0	13.5001	True	11
12	exact	20/29	600	0.0138	0.0090	12.6025	True	12
13	direct	20/29	600	0.0129	0	13.4929	True	13
14	exact	20/47	600	0.0320	0.0114	12.9103	False	14
15	direct	20/47	600	0.0170	0	13.6205	True	15
16	product	10/11	600	0.0163	0.0052	3.9801	True	16
17	single	10/11	600	0.0162	0.0052	3.9075	True	17
18	product	10/29	600	0.0187	0.0035	3.9346	True	18
19	single	10/29	600	0.0186	0.0034	3.9263	True	19
20	product	10/47	600	0.0141	0.0050	3.9233	True	20
21	single	10/47	600	0.0140	0.0050	3.9264	True	21
22	exact*	10/11	1000	0.0236	0.0019	11.1414	True	22
23	direct*	10/11	1000	0.0278	0	12.4295	False	23
24	exact*	10/29	1000	0.0249	0.0022	10.9333	True	24
25	direct*	10/29	1000	0.0276	0	12.3312	False	25
26	exact*	10/47	1000	0.0281	0.0026	11.0150	False	26
27	direct*	10/47	1000	0.0293	0	12.3768	False	27
28	exact*	20/11	1000	0.0200	0.0040	20.8834	True	28
29	direct*	20/11	1000	0.0211	0	22.9271	True	29
30	exact*	20/29	1000	0.0200	0.0034	20.0811	True	30
31	direct*	20/29	1000	0.0227	0	21.9897	True	31
32	exact*	20/47	1000	0.0216	0.0071	19.9588	True	32
	direct*	20/47	1000	0.0244	0	21.7639	True	

An asterisk marks the follow-up fitting domain $[-1.5, .5]^d$ and 1,000 updates.
 Product uses two independent two-probe banks; single squares a four-probe trace.

Every row retains its own history and exact-trace diagnostic set. Runtime is not a hardware-independent complexity bound.

12.4. Policy-payoff diagnostics

Method	d/seed	Critic at zero	Euler mean	MC s.e.	Critic minus mean
exact	10/11	-1.1925	-1.3045	0.0041	0.1120
direct	10/11	-1.2054	-1.2729	0.0036	0.0675
exact	10/29	-1.1781	-1.2999	0.0040	0.1218
direct	10/29	-1.1899	-1.2701	0.0036	0.0802
exact	10/47	-1.2080	-1.2967	0.0040	0.0887
direct	10/47	-1.2077	-1.2698	0.0036	0.0621
exact	20/11	-1.1809	-1.3458	0.0041	0.1649
direct	20/11	-1.1986	-1.2834	0.0035	0.0848
exact	20/29	-1.1920	-1.2895	0.0035	0.0975
direct	20/29	-1.2071	-1.2754	0.0034	0.0684
exact	20/47	-1.0343	-1.5052	0.0036	0.4709
direct	20/47	-1.1661	-1.3051	0.0036	0.1390
exact*	10/11	-1.2974	-1.2521	0.0035	-0.0453
direct*	10/11	-1.2992	-1.2527	0.0035	-0.0465
exact*	10/29	-1.2933	-1.2521	0.0035	-0.0412
direct*	10/29	-1.2871	-1.2521	0.0035	-0.0350
exact*	10/47	-1.2939	-1.2527	0.0035	-0.0412
direct*	10/47	-1.2959	-1.2527	0.0035	-0.0433
exact*	20/11	-1.2788	-1.2646	0.0034	-0.0142
direct*	20/11	-1.2787	-1.2656	0.0034	-0.0132
exact*	20/29	-1.2712	-1.2646	0.0034	-0.0066
direct*	20/29	-1.2563	-1.2689	0.0034	0.0126
exact*	20/47	-1.2772	-1.2645	0.0034	-0.0127
direct*	20/47	-1.2821	-1.2652	0.0034	-0.0169

These estimates use 2,048 paths and 80 Euler steps from zero log capital. Forty-step samples are also saved. Identical random seeds pair policy comparisons at the same step size; the forty- and eighty-step simulations are not a Brownian-coupled

1	refinement estimator. The standard error measures sampling error only. The initial	1
2	critic is not an external optimal-value reference.	2
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31		31
32		32